HARD CODERS

Engineering with AI

Blackbeard’s Bounty

30+	6×5	96%	Codex
Prompts	Grid	RTP Target	Tool

HARD CODERS · TEAM 13

OpenAI Codex · HTML / CSS / JS · Team of 10 · April 2026

"No shortcuts. We build on the foundation."

Research & Planning

Domain Research

- 10 real slot games analyzed by team members
- RTP math, volatility & scatter-pay systems
- Tumble/cascade mechanic (Sweet Bonanza model)
- Legal rules for entertainment vs. gambling apps
- Mobile UX patterns, portrait mode, one-hand play
- User reviews: what players love and hate

User Research

6
Personas

6
User Stories

User Types Covered

- Casual players (Sarah, Alex, Mia)
- Power/progression players (Michael, Donald)
- Analytical players (Robert)

Research grounded every AI prompt that worked. Without domain knowledge, we would have prompted blindly.

Personas & User Stories

Sarah, 22
Casual Student

Needs: Easy controls, quick sessions, no login required

"Start playing immediately without creating an account"

Michael, 34
Progression Seeker

Needs: Achievements, multiple themes, stats tracking

"Unlock new themes and achievements over time"

Donald, 46
Power Player

Needs: Auto-spin 100+ rounds, high volatility, big wins

"Set auto-spin and keep the game running hands-free"

Robert, 78
Analytical Player

Needs: Low stakes, win/loss dashboard, minimalist UI

"Access real-time RTP stats and spin history"

Alex, 20
Break-Time Player

Needs: Simple interface, clear buttons, instant results

"See win or loss clearly after every spin"

Mia, 22
First-Time User

Needs: Beginner-friendly design, no jargon, clear feedback

"Learn how to play quickly with clear instructions"

AI Process & Final Product

1

Baseline

1×3 working game
in 2 prompts

2

Scale Up

6×5 grid + tumble
win mechanic

3

Polish

Theme, icons,
coin animations

Features Delivered by Hard Coders

- 6×5 Scatter Grid
- Pirate Themed Custom Icons
- Easy & Hard Difficulty Modes
- Rules Popup for New Players
- Column by Column Spin Animation
- Tumble / Cascade Mechanic
- Coin Rain Animations on Wins
- Light / Dark Mode Toggle
- 96% Target RTP Calibration
- Spin Lock (no spamming)

Key Findings

🎯 Narrow prompts win

Every broad goal ('make it fun') produced inconsistent results. Every specific, scoped prompt ('add a clickable rules popup') succeeded in 1-2 attempts.

🔍 AI can't replace human debugging

There was a hidden bug in the lever that couldn’t be fixed by any AI attempt. One HardCoder using browser DevTools found it in minutes. AI was able to fix it once given the exact location.

📚 Domain knowledge drives prompts

Our research paid off directly. Knowing what tumble mechanics, RTP, and scatter-pays meant let us give the AI the correct specs instead of just vague requests.

🔄 Iteration is the process

The coin animation alone took 5 prompts. The lever took 7+ before we pivoted. Prompt iteration is real development work, we need to plan for it in our timeline.

Meeting SE Standards

Modularity

★★★★★

Structured across plan, source, and assets with JS split into config, slotMachine, ui, main

Linting

★★★★☆

ESLint + W3C HTML validation run after each generation pass

Documentation

★★★★☆

JSDoc annotations + named constants throughout config.js

Testing

★★★☆☆

Unit tests for win logic,
RTP simulation, tumble mechanic

Clean Code

★★★★★

Descriptive names, small fns,
DRY config, error handling

Repo Hygiene

★★★★☆

Frequent commits by HardCoders,
.gitignore, clean file structure

Conclusion: Should We Use AI?

✓ AI Is Useful For

- Rapidly generating a working baseline
- Implementing scoped, isolated features
- Boilerplate & repetitive code (tests, configs)
- Fixing specific, named bugs with location info
- Documentation scaffolding

✗ AI Is NOT Reliable For

- Self-directed architecture at scale
- Visual UX polish without actual quantified goals
- Finding cross-file bugs without location info
- Consistency without linting & human review
- Replacing engineering judgment

Hard Coders verdict: Treat AI like a capable junior dev, review everything, one task at a time, keep your judgment active.

BLACKBEARD'S BOUNTY

Switch to Hard Mode

Switch to Light Mode

Rules

Mode: **Easy**

Balance: **$300.00**

Fixed Bet: **$10.00**

Board Size: **6 x 5**

Easy mode active: $10 spins with steadier, lower-volatility payouts.

SPIN